

STABLE: Semantic Tracking and Backbone-Linked Environment for XR Teleoperation

Abstract—Teleoperating a mobile robot in an unknown environment is often challenging because operators can only rely on a limited camera field of view. While dense 3D reconstructions can improve spatial awareness, a tight integration may come with substantial computational and memory requirements. Object-level representations provide complementary semantic information in a more compact form. We present STABLE, a semantic tracking and backbone-linked environment designed for XR-based robot teleoperation. This work combines instance-level semantic mapping with a LiDAR SLAM backbone, allowing objects and geometry to be represented in the same map frame. Object geometries are estimated from LiDAR points associated with instance segmentation masks and are added to the map only after their estimates have stabilised, providing a consistent view for the teleoperator. The proposed system is demonstrated qualitatively on a Husky robot in an indoor environment setup.

I. INTRODUCTION

A teleoperator controlling a ground robot in an unknown environment often relies mainly on a forward-facing camera. This limited view reduces spatial awareness and makes depth hard to judge, which immersive three-dimensional reconstructions of the remote scene can address. Previous studies have shown that such displays can improve navigation accuracy and situational awareness compared with video-only teleoperation [1].

Geometric reconstructions describe the shape and layout of the environment, but they do not always provide object-level information. For teleoperation, this information is important because the operator must distinguish between objects such as people, obstacles, and landmarks. Semantic mapping supplies this context, but is rarely integrated with human-facing teleoperation, while extended-reality teleoperation systems remain largely geometric.

This paper brings these two areas together. An instance-level semantic layer operates alongside the LiDAR SLAM backbone and uses the same optimised map frame. The resulting map, which combines geometry with object information, is presented to the teleoperator through an immersive display. Because the consumer is a human operator, objects are shown only after their estimates have converged.

With this paper we present two main contributions:

- *An object-level semantic map built for a human in the loop.* An instance-level semantic layer runs alongside a LiDAR SLAM backbone and is coupled to it by a single quantity, the camera pose in the optimised map frame, so that objects and geometry are co-registered by construction rather than by a separate alignment step. The resulting map is delivered to a teleoperator

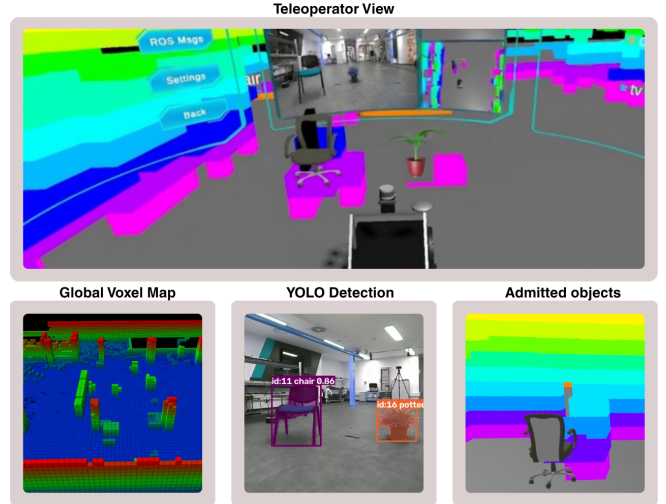


Fig. 1: System overview. The teleoperator (left) sees admitted objects as scaled class proxies within the LiDAR map. The physical scene (top right) is mapped by the SLAM backbone (bottom left) while instance segmentation (bottom centre) selects the LiDAR returns from which each object’s geometry is estimated; objects are rendered only after admission (bottom right).

in an immersive display, closing the path from onboard perception to the operator who acts on it.

- *Stability treated as a requirement of the representation.* A planner tolerates an object whose estimate keeps changing; a human watching an immersive display does not, and would rather wait for a stable one. We make this explicit as an admission criterion that withholds an object until its estimate has converged and treat the resulting delay as a deliberate cost.

II. RELATED WORK

Semantic mapping methods mainly differ in how they associate semantic information with geometric maps. Dense methods assign a semantic class to every geometric element, such as a surfel or voxel. For example, SuMa++ [2] combines semantic labels with a LiDAR-based surfel map and uses them to identify and remove dynamic objects. Similar approaches have also been developed for volumetric map representations [3]. These methods classify every point but do not individuate objects, which is what a teleoperator reasons about.

To address this limitation, object-level mapping methods explicitly represent individual object instances. These objects

can be stored as landmarks in a state estimator or as nodes in a hierarchical scene graph [4], [5]. More recent methods also support open vocabulary object categories by incorporating vision-language features [6]. In this work, we adopt the object-level representation with a closed set of object categories and a simple scene model. This suits an operator who needs identity, position, and size rather than inter-object relations.

Camera categories and LiDAR geometry are commonly fused by projecting the cloud into the image and associating returns with a detection region. Frustum PointNets [7] established the pattern for 2D boxes, and instance masks refine it by replacing the box with the object silhouette, which admits far fewer background returns. Closest to this work is Artifacts Mapping [8], which fuses camera detections with depth to localise objects in 3D and register them in a semantic map.

Recovering object geometry from sparse, single sided observations is underconstrained, and object-level SLAM has consistently addressed this with category-level priors. SLAM++ [9] takes the strongest form, matching detections against a database of known CAD models. Later work relaxes this to compact parametric shapes: CubeSLAM [10] represents objects as cuboids and QuadricSLAM [11] as constrained dual quadrics, with extensions introducing explicit object size priors [12] or learned category specific shape embeddings [13].

Immersing the operator in a reconstruction of the remote scene addresses the degraded spatial awareness of video-based teleoperation directly: Stotko et al. [1] stream a live SLAM reconstruction to a VR client and report, in a user study, higher situation awareness and more precise navigation than a purely video-based baseline. Surveys of virtual and mixed reality in human-robot interaction document a wide design space of such interfaces [14]. These systems are almost uniformly geometric, however, presenting the operator with reconstructed surfaces rather than with identified objects, and conversely, the semantic mapping literature rarely closes the loop to a human consumer at all.

III. METHOD

Figure 2 summarises the system. We describe the geometric backbone only to the extent that the semantic layer depends on it (section III-A), then mask-conditioned object measurement (section III-B), recursive estimation under a class-conditioned size prior (section III-C), and the convergence criterion governing admission to the map (section III-D).

A. Geometric Backbone and Frame Convention

The geometric layer is a multi-level scan matching LiDAR SLAM system with submap based loop closure and incremental pose graph optimisation [15]. The relevant frames are the LiDAR l , the camera optical frame c , the robot base b , the odometry frame o , and the optimised map frame m .

The semantic layer consumes a single quantity from the backbone: the transform ${}^m\mathbf{T}_c$ from the camera optical frame to the optimised map frame, obtained from the transform

tree as ${}^m\mathbf{T}_c = {}^m\mathbf{T}_o {}^o\mathbf{T}_b {}^b\mathbf{T}_c$. The backbone contributes the correction ${}^m\mathbf{T}_o$ produced by the pose graph; ${}^o\mathbf{T}_b$ is the platform odometry; and ${}^b\mathbf{T}_c$ is the fixed kinematic chain from the robot base to the camera given by the platform model. Object measurements are lifted into m by this transform, placing them in the same frame as the geometric map presented to the operator.

The scope of the object layer is deliberately local. Objects not reobserved within a fixed timeout are pruned, so the published set describes the robot’s immediate surroundings rather than a persistent global map, and no object state is written back into the global map. Persisting objects into the global map, which would require correcting them under loop closure, is left to future work.

B. Mask-Conditioned Object Measurement

Each image is segmented into instance masks $\mathcal{M}_i$ with persistent identities τ_i and classes c_i . Detections are discarded if they fall outside the class set of interest, have confidence below 0.45, receive no tracker identity, or lie within 5 px of the image border, since truncation systematically biases the extent estimate.

The point cloud is transformed into the camera optical frame using ${}^c\mathbf{T}_l$ interpolated at the cloud timestamp and projected through the pinhole model, retaining points that fall inside the image and within 30 m. Each mask then acts as a viewing volume, selecting the subset $\mathcal{S}_i$ of projected points for which $\mathcal{M}_i(\mathbf{u}_k) > 0.5$. This subset is contaminated, since background surfaces visible through the object silhouette project inside the same mask; taking the 20th percentile of the selected depths as a reference biased towards the near surface, we retain only points within $\delta = 0.40$ m of it and reject the detection when fewer than $n_{\min} = 8$ returns survive.

With $\mathbf{q}^-$ and $\mathbf{q}^+$ the componentwise 5th and 95th percentiles of the retained points, the measurement is

$$\mathbf{z}_i = \left({}^m\mathbf{T}_c \frac{\mathbf{q}^- + \mathbf{q}^+}{2}, \mathbf{d}_i \right), \quad (1)$$

where $\mathbf{d}_i$ collects the per-axis spans $\mathbf{q}^+ - \mathbf{q}^-$, reordered from the optical convention to width, depth and height. These spans are axis-aligned in the camera frame and therefore overestimate the extents of an obliquely viewed object; gross cases are rejected by the admission gate of Sec. III-C, but a moderate overestimate is retained, and recovering extents in the object frame is left to future work.

C. Recursive Object Estimation

Detections are associated across frames by the identity τ_i carried by the segmentation, rather than by nearest neighbour search in the map frame, which becomes ambiguous whenever several instances of one class stand close together.

Position is the component-wise running median of a bounded observation history $\mathcal{H}_i$. The median absorbs the isolated error produced by a momentarily poor mask, but over a short window it will eventually follow a sustained one, so an observation lying farther than $r_{\text{out}} = 2.0$ m from the

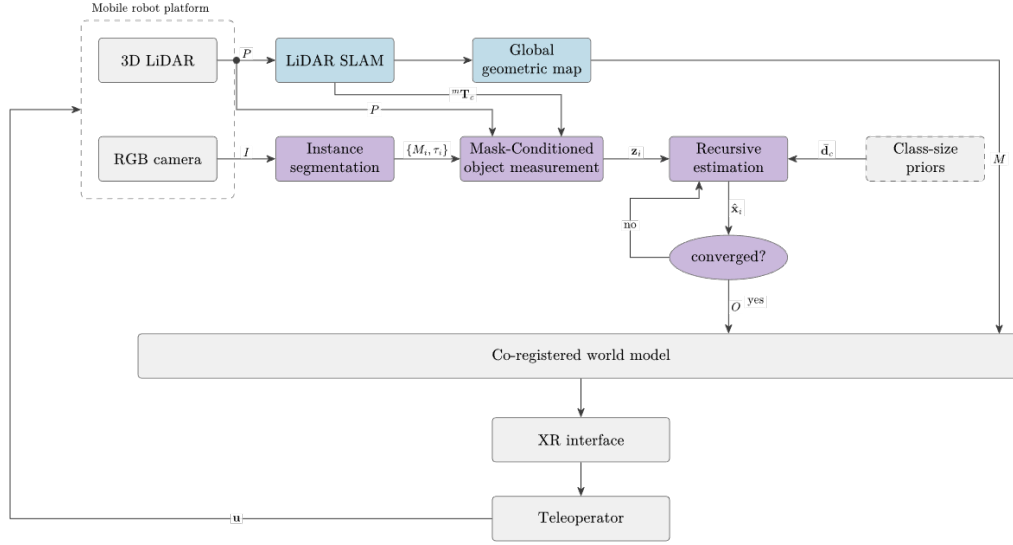


Fig. 2: System architecture. The LiDAR SLAM backbone (blue) and the instance-level semantic layer (violet) run in parallel on the same platform and are coupled by the camera pose in the optimised map frame, so that objects and geometry are co-registered by construction. Objects are admitted to the map only once their estimates have converged, and the resulting world model is rendered to the teleoperator, who closes the loop with velocity commands.

running mean is rejected before the history is updated. Such an observation is discarded in its entirety rather than merely withheld, since a gross positional error indicates a mask that cannot be trusted for the extent measurement either.

Object extent is particularly difficult to estimate from sparse LiDAR data. A 16-beam LiDAR produces only a small number of returns on distant objects, so the object dimensions may be poorly constrained during the first few observations. We therefore use the nominal category extent $\bar{\mathbf{d}}_c$ only for initialisation and measurement validation, rather than as a permanent constraint on the estimate. Each object is initially assigned the nominal extent of its class. A new extent measurement is accepted only if it remains within a fixed ratio of the nominal extent along all three axes:

$$0.3 = \rho_{lo} \leq \frac{[\mathbf{d}_i]_a}{[\bar{\mathbf{d}}_{c_i}]_a} \leq \rho_{hi} = 3.0, \quad a \in \{1, 2, 3\}, \quad (2)$$

This test rejects unrealistic measurements, such as cases where segmentation errors cause the estimated box to include part of the background. After $n_d = 3$ valid measurements have been collected, the reported extent is computed as their component-wise median. The nominal category extent then no longer affects the estimate, although it continues to be used for rejecting implausible measurements. In this way, the prior supports the estimate when the observations are insufficient and gives way to the measured data once enough evidence is available.

D. Convergence and Admission to the Map

An object is admitted to the published map i.e. to the teleoperator’s view only once its position estimate has stabilised. Position is declared settled when at least $n_p = 8$ observations

have accumulated in $\mathcal{H}_i$ and the per axis standard deviation is below $\sigma_p = 0.15$ m. The estimate is then held fixed: further observations of the track refresh its timestamp, and so keep it alive, but no longer alter its position. Large changes in an object’s estimated position between updates cause visible jitter, reducing the user’s confidence in the entire scene. Although the gate delays the appearance of newly detected objects until their estimates have converged, we consider this an acceptable tradeoff, as a delayed but stable object provides a better user experience than one that appears unstable.

Tracks not re-observed within $T = 8$ s are pruned, which bounds the published set to the robot’s immediate surroundings. Tracks of the same class whose estimates fall within a merge radius $r_m = 1.0$ m are combined, suppressing the duplicates that arise when the segmentation assigns a fresh identity to an object that leaves and re-enters the field of view. The published set $\mathcal{O}$ therefore contains only converged objects, each carrying a class, a persistent identity, a position, and metric extents.

IV. IMPLEMENTATION AND QUALITATIVE RESULTS

A. Setup

The system runs on a Clearpath Husky A200 carrying a Velodyne VLP-16 and a forward facing RGB-D camera, and in an Ignition Gazebo model of the same platform in an indoor environment containing different objects. All components run on the Robot PC (Nvidia Jetson Orin Nano), the operator who is wearing a HTC Vive Pro headset, sees the unity application interface running on another PC. Instance segmentation uses YOLO11-seg with a multi-object tracker for identity association, and the geometric backbone is the

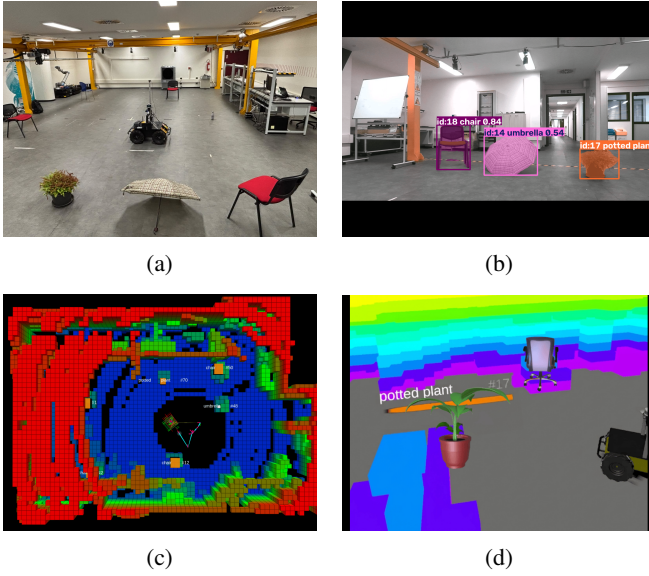


Fig. 3: The system operating in an indoor laboratory. (a) The physical scene. (b) Instance segmentation on the camera image, from which the LiDAR returns belonging to each object are selected. (c) The global voxel map produced by the SLAM backbone, with the detected objects and the robot trajectory shown in the same optimised frame. (d) The teleoperator's view, in which each admitted object appears as a class-specific proxy scaled to its estimated extents and annotated with its class and persistent identity.

LiDAR SLAM system of [15] in its default configuration. All thresholds reported above were fixed across runs.

B. Qualitative Behaviour

The results presented in Fig. 3 are qualitative. They are intended to demonstrate that the proposed representation is constructed as described and is usable by a human operator, rather than to quantify its accuracy. A quantitative evaluation against simulator ground truth is the immediate next step and is discussed in Sec. V.

Fig. 3a shows the experimental setup. Fig. 3b illustrates the instance segmentation output, where the detected object masks are used to generate object representations in the optimised map frame. These objects are visualised in the RViz view shown in Fig. 3c. Fig. 3d presents the operator's view, where objects are admitted only after they have been consistently observed for eight consecutive frames. Pedestrians are detected and tracked, but because they do not satisfy the position spread criterion while moving, they are not admitted to the map. This follows directly from the admission rule based on the stability of the position estimate, as discussed further in Sec. V.

V. DISCUSSION & CONCLUSION

We have described a system that couples an instance-level semantic layer to a LiDAR SLAM backbone through a single shared quantity, the camera pose in the optimised map frame, and delivers the resulting object-annotated map

to a teleoperator in an immersive display. The design is shaped by that consumer throughout: geometry is recovered from sparse returns under a category level prior that is used to initialise and validate rather than to substitute for measurement, and objects are withheld from the operator until their estimates have stabilised. The system has not yet been evaluated quantitatively. Measuring object position and extent error against simulator ground truth, together with the interval between first detection and admission, is the immediate next step and is required before any accuracy claim can be made.

REFERENCES

- [1] P. Stotko, S. Krumpen, M. Schwarz, C. Lenz, S. Behnke, R. Klein, and M. Weinmann, "A VR system for immersive teleoperation and live exploration with a mobile robot," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 3630–3637.
- [2] X. Chen, A. Milioto, E. Palazzolo, P. Giguère, J. Behley, and C. Stachniss, "SuMa++: Efficient LiDAR-based semantic SLAM," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 4530–4537.
- [3] A. Rosinol, M. Abate, Y. Chang, and L. Carlone, "Kimera: an open-source library for real-time metric-semantic localization and mapping," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 1689–1696.
- [4] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," in *Robotics: Science and Systems (RSS)*, 2022.
- [5] M. Grinvald, F. Furrer, T. Novkovic, J. J. Chung, C. Cadena, R. Siegwart, and J. Nieto, "Volumetric instance-aware semantic mapping and 3d object discovery," *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 3037–3044, 2019.
- [6] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 5021–5028.
- [7] C. R. Qi, W. Liu, C. Wu, H. Su, and L. J. Guibas, "Frustum pointnets for 3d object detection from rgb-d data," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 918–927.
- [8] F. Rollo, G. Raiola, A. Zunino, N. Tsagarakis, and A. Ajoudani, "Artifacts mapping: Multi-modal semantic mapping for object detection and 3D localization," in *European Conference on Mobile Robots (ECMR)*. IEEE, 2023, pp. 1–8.
- [9] R. F. Salas-Moreno, R. A. Newcombe, H. Strasdat, P. H. Kelly, and A. J. Davison, "Slam++: Simultaneous localisation and mapping at the level of objects," in *2013 IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 1352–1359.
- [10] S. Yang and S. Scherer, "Cubeslam: Monocular 3-d object slam," *IEEE Transactions on Robotics*, vol. 35, no. 4, pp. 925–938, 2019.
- [11] L. Nicholson, M. Milford, and N. Stunderhauf, "Quadricslam: Dual quadrics from object detections as landmarks in object-oriented slam," *IEEE Robotics and Automation Letters*, vol. 4, no. 1, pp. 1–8, 2019.
- [12] K. Ok, K. Liu, K. Frey, J. P. How, and N. Roy, "Robust object-based slam for high-speed autonomous navigation," in *2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 669–675.
- [13] J. Wang, M. Rünz, and L. Agapito, "Dsp-slam: Object oriented slam with deep shape priors," in *2021 International Conference on 3D Vision (3DV)*, 2021, pp. 1362–1371.
- [14] M. E. Walker, T. Phung, T. Chakraborti, T. Williams, and D. Szafir, "Virtual, augmented, and mixed reality for human-robot interaction: A survey and virtual design element taxonomy," *ACM Transactions on Human-Robot Interaction*, vol. 12, no. 4, 2023.
- [15] F. Rollo, V. Pericu, M. Roveri, A. Ajoudani, and N. Kashiri, "LEO-SLAM: A multi-level scan matching approach with submap-based loop closure detection," in *European Conference on Mobile Robots (ECMR)*, 2025.